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A culturomics approach to quantifying the salience of species on the global internet

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Abstract

1. Conservation will only be successful over the long term if people support conservation goals. While many factors may influence the level of such support, it is clear that people are more willing to conserve species and places that they are familiar with and which provide them with something they value.
2. Until now this dimension has been largely lacking from conservation decision-making and its underlying scientific evidence base. This is understandable given conservation scientists' historic focus on population and community ecology and the practical difficulties associated with assessing the cultural prominence of species or places at anything more than local scale.
3. This latter challenge is rapidly being addressed through a new generation of culturomic metrics that takes advantage of publicly available digital content.
4. Here, we suggest that one such metric, estimated frequency of webpages that mention the scientific names of a species, broadly reflects the relative prominence of a species in global culture.
5. Using all extant bird species as a case study, we demonstrate that species that are mentioned at high frequency on the global internet: (a) were scientifically described earlier, (b) have wide geographic ranges that overlap with technologically advanced societies, (c) are phenotypically conspicuous and (d) are strongly associated with direct human interactions (e.g. hunting).
6. These results support the use of estimates of scientific name frequency on the internet as a proxy of a species' cultural salience, especially with respect to urban populations in technologically advanced societies.
7. We conclude by discussing how a new generation of digital tools might be developed to support conservation monitoring and communication.

KEYWORDS

birds, cultural value, culturomics, digital methods, human–nature interactions

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1 | INTRODUCTION

One of the most conspicuous trends in 21st century conservation theory and practice is its increasingly people-centric approach (Holmes, Sandbrook, & Fisher, 2017; Kareiva & Marvier, 2012). However, the geographies of intervention, discourses and institutions of contemporary conservation are still largely shaped by earlier paradigms (Caughley, 1994; Soulé, 1985), with metrics and classifications largely derived from population and community ecology (e.g. Red Lists, habitat classifications) forming the basis of contemporary legislation and priority setting schemes (e.g. hotspots, Key Biodiversity Areas, etc.). Developing and introducing metrics of nature–people interactions is therefore a key strategy to support innovative new approaches and help revitalize and re-orientate conservation institutions (Ladle et al., 2016).

Biological species clearly differ in their cultural profile and importance across societies, with consequences for conservation decision-making. For example, conservationists have long mobilized well-known charismatic species as a tool for generating public support, influencing policy and promoting conservation values (Bowen-Jones & Entwistle, 2002; Home, Keller, Nagel, Bauer, & Hunziker, 2009). Moreover, the presence of culturally prominent (salient) species in an area may have a safeguarding effect through their influence on 'rallying power', that is, the public are more likely to support campaigns against developments that threaten the populations and habitats of familiar species that engender positive sentiment (Brambilla, Gustin, & Celada, 2013). Until recently, quantifying the prominence of multiple species within and between cultures required costly social surveys that were necessarily limited in geographic and taxonomic scope. While such surveys provide excellent contextual information and exceedingly detailed information on cultural value, they cannot be easily scaled up or compared due to the geographies of cultural diversity and a general lack of systematization in social survey methodologies (Bragagnolo, Malhado, Jepson, & Ladle, 2016).

The global internet, with its high levels of social participation, offers an alternative source of information on the cultural salience of species. Such information can be collected, analysed and interpreted through culturomics, an emerging area of study that explores human culture through the quantitative analysis of word frequencies in large digital bodies of text (Michel et al., 2011). It is based on the idea that words are symbolic representations of concepts, places or objects (Carlston, 2013) and, by extension, the frequency with which they appear within a digital corpora (such as searchable web pages) provides information about the cultural salience of the thing they symbolize. Culturomics has enormous potential to provide new insights for conservation in general and people–nature relationships in particular, passively crowdsourcing the sedimented histories of cultural engagements (direct and indirect) with species and locating the tone and mood of contemporary interactions with them (Ladle et al., 2016).

One of the biggest potential limitations of culturomic analysis is that the same words may have more than one meaning (homonyms). For example, teal (*Anas carolinensis*) is an attractive small duck whose name is also used for a popular colour of paint. Additionally, the same biological species may have various common names both within and between languages. These limitations pose serious challenges for global or regional scale analysis and the first culturomic analyses of the internet representation of species were consequently performed within a single language group (Correia, Jepson, Malhado, & Ladle, 2016; Schuetz, Soykan, Distler, & Langham, 2015). The use of scientific binomial names of species potentially avoids the homonym problem because Latin is a 'dead' language and the scientific name typically applies only to the biological organism. However, the frequency of scientific name use may not accurately reflect the representation of a species on the internet if only specialists use it. Two recent studies strongly suggest that this is not the case by demonstrating that scientific name frequency is highly correlated with common name frequency over a range of digital corpora. Specifically, Jarić, Courchamp, Gessner, and Roberts (2016) found strong correlations between the frequency of Latin and common names of selected species of diurnal birds of prey, Carnivora, Primates and marine mammals on internet pages, social media networks (Twitter and Facebook) and newspapers. Similarly, Correia, Jepson, Malhado, and Ladle (2017) demonstrated that strong and consistent correlations exist between the use of vernacular and scientific names of bird species on the internet at both global and country level and, significantly, that the correlation persisted independent of country's native language. These studies strongly suggest that scientific name frequency can be used as a good general proxy of vernacular name frequency in culturomic analysis, paving the way for a global analysis.

Capitalizing on these recent breakthroughs, we quantify the representation of extant bird species on the global internet using search engine-derived estimates of the frequency of webpages that mention their scientific name. We then demonstrate that highly represented species on the global internet are statistically associated with characteristics that are predicted to contribute to their cultural prominence (salience) in the real world, especially among urban populations in technologically advanced societies.

2 | WHAT DRIVES THE DIFFERENTIAL REPRESENTATION OF SPECIES ON THE INTERNET?

We reasoned that people are most likely to generate internet content about species that are prominent in their mental models of the world, a characteristic we term 'cultural salience'. Cultural salience of a species can be conceptualized as an emergent property of the history of human interactions (both positive and negative) with wild/captive individuals, derived products (e.g. skins, meat) and/or representations (words, images, videos). These interactions emerge when a species *affords* (sensu Gibson, 2014) purpose, actions and emotions to humans. Empirical studies suggest that certain attributes of species afford more to humans than others. For example, species with large body size afford more meat or heighten prestige to

hunters and are popular exhibits for zoos (Frynta, Šimková, Lišková, & Landová, 2013; Ward, Mosberger, Kistler, & Fischer, 1998), species that have phenotypic similarity with humans (Batt, 2009) and/or paedomorphic/infantile features (Borgi & Cirulli, 2016, 2013; Bradshaw & Paul, 2010) afford feelings of compassion, 'cuteness', and kinship, and species perceived as intelligent, beautiful and/or singular afford aesthetic and scientific contemplation (Ducarme, Luque, & Courchamp, 2013).

Given this, we posit that the cultural salience of a species will depend on both what species afford to humans and the frequency of interaction between species (or their representations) and people. It is important to note that, because affordances are 'triggered' by interactions (Gibson, 2014), the distribution and abundance of species both in nature and in the representational realm is predicted to influence cultural salience. Moreover, the more culturally salient a species becomes, the more likely it is that it will be represented, further increasing its salience among individuals who encounter those representations. Such a positive feedback loop loosens the link between living populations and their representation in global culture, potentially leading to a disconnection between public perceptions of species abundances and their natural populations (Courchamp et al., 2018). This disconnection may be especially prominent among urban populations in advanced societies who have few interactions with species in natural settings. Positive feedback between representations and salience may also be responsible for the high cultural prominence of some extinct species such as the dodo (*Raphus cucullatus*; Turvey & Cheke, 2008), or even mythical species such as the yeti, which can maintain or increase their cultural salience in the absence of an extant population.

In addition to distributional range and abundance, encounters with non-human species in nature will be influenced by phenotypic traits (e.g. large body size, conspicuous coloration, diurnal or confiding behaviour) that influence the ease with which a species can be found and recognized when present in the environment. Other more contextual factors, such as local cultural prominence, may increase/decrease the chance that a species will be detected (Gosler, 2017; Hunn, 1999). There is good empirical evidence of the link between human encounters with species in nature and their cultural salience. A recent study from the US observed that Google searches (as measured by relative search volume using Google Trends) for the common names of 68 resident bird species were positively associated with regional estimates of bird population densities (Schuetz et al., 2015). Similarly, the most important factor driving the internet representation of Brazilian bird species on Brazilian web sites was the estimated human population size within the birds' geographical distribution in the country (Correia et al., 2016).

In summary, we argue that species cultural salience can be thought of as a relational property, delimited by the species' intrinsic traits (e.g. size, behaviour, edibility) and its interplay and interactions with human culture (e.g. frames, narratives, practices, technologies; Jepson & Barua, 2015) and psychology (Frynta et al., 2011, 2013; Herzog Jr & Burghardt, 1988; Lišková & Frynta, 2013; Lišková, Landová, & Frynta, 2015).

3 | MATERIALS AND METHODS

We use internet salience (i.e. the frequency of species representation on the internet) of a species as a broad proxy of its cultural salience. Our core metric of species representation on the global internet is the common logarithm (log10) of the estimated number of webpages that mention the scientific (binomial) name of each of the world's extant bird species. Webpage frequency was estimated with an automated and depersonalized web search using the Google Custom Search API. Scientific names of all extant bird species (n = 10,944) were taken from the Birdlife/IUCN taxonomic checklist (Birdlife International, 2010). To account for changes in scientific names due to splitting, clumping and temporary loss of full species status we: (a) included common taxonomic synonyms in our internet representation index (Correia et al., 2018) and, (b) assessed nomenclatural stability of each species based on the co-occurrence of a species' name on six global bird species checklists (Birdlife International, 2010; Del Hoyo, Elliott, & Sargatal, 2013; Gill & Donsker, 2018; Peters, 1987; Sibley & Monroe, 1990). We considered a species as nomenclaturally stable if the same scientific name was used on all six lists.

Our explanatory variables (full description in Table S3) represent geographical, cultural, phenotypic and ecological factors that are predicted to influence the global cultural salience of species and, by extension, their representation on the global internet. Geographical traits are predicted to be particularly important since they will be associated with both familiarity with a species and the capacity of the human population that is engaging with the species to contribute internet content. The underlying logic being that, other conditions being equal, societies and cultures with a higher frequency of internet users will be more likely to digitally document their encounters with non-human species. Thus, in addition to geographic factors that may influence the frequency of human engagements with species (range size, presence in urban environments), we also calculated an internet capacity (IC) index for each species based on the percentage of global internet users in the countries where the species is present. A full list of candidate explanatory variables, information sources and justification for inclusion in the model is provided in Table S3.

We adopted a multi-model inference approach and calculated all possible model combinations (without interactions) relating internet salience to our set of explanatory variables using generalized linear models with Gaussian error distribution implemented through the MuMIn package for R Software. Models were run separately for nomenclaturally stable species only and for all species irrespective of recent taxonomic history. We then identified the best performing models according to Akaike's Information Criterion corrected for sample size (AICc) and Akaike's weights (ωAICc). All plausible models showed high explanatory power (adjusted r^2 > .5), particularly when accounting for taxonomic stability (Tables S4–S6). However, because no single model clearly outperformed the others, we used a model averaging approach to obtain averaged parameter estimates and the relative importance of each explanatory variable (Tables S2 and S3). For this process, we considered only models with ΔAICc ≤ 10 as a plausible fit for the observed data, given the candidate set of models

(Burnham & Anderson, 2004, 2003). All analyses were implemented in R Software v3.1.3 (R Core Team, 2015).

4 | RESULTS

Only a small proportion of the 10,944 extant bird species have high internet salience (estimated frequency of webpages that mention their scientific name). Different forms of human engagement are strongly associated with high internet salience (Figure 1, Tables S4–S6). Specifically, factors associated with opportunities to detect (e.g. range size, body size, nocturnal behaviour), interact (game species, cage species) and report (web capacity) engagements with species have a strong positive effect on cultural representation (Figure 1a–f).

The strongest association was with the number of years since a species was first scientifically named (Figure 1a). Bird species used in aviculture, kept as pets (Figure 1b) or hunted as game (Figure 1c) also have very high internet salience (Figure 1c). Species that flourish in human-dominated landscapes tended to have higher internet representation than those restricted to natural habitats, reflected in the high salience of bird species found in urban environments (Figure 1e) and species with expanding populations (Figure 1f). Higher endangerment categories are also generally associated with higher internet representation (Table S4).

5 | DISCUSSION

5.1 | Validity and robustness of metric of internet salience

Our results broadly support our proposition that the estimated frequency of webpages on the global internet that mention the scientific names of species (internet salience) can be considered as a broad proxy of the global cultural salience of species, being universally applicable, highly specific, replicable (within limits) and time-dependent. However, as with any new area of investigation, our culturomics approach has various limitations and assumptions that need to be addressed or taken into account when interpreting our results and evaluating the viability and reliability of our proposed internet representation metric (Kitchin, 2014). Some limitations and biases are inherent to our methods of data capture/analysis and to language dynamics, while others are due to the absence of datasets on species

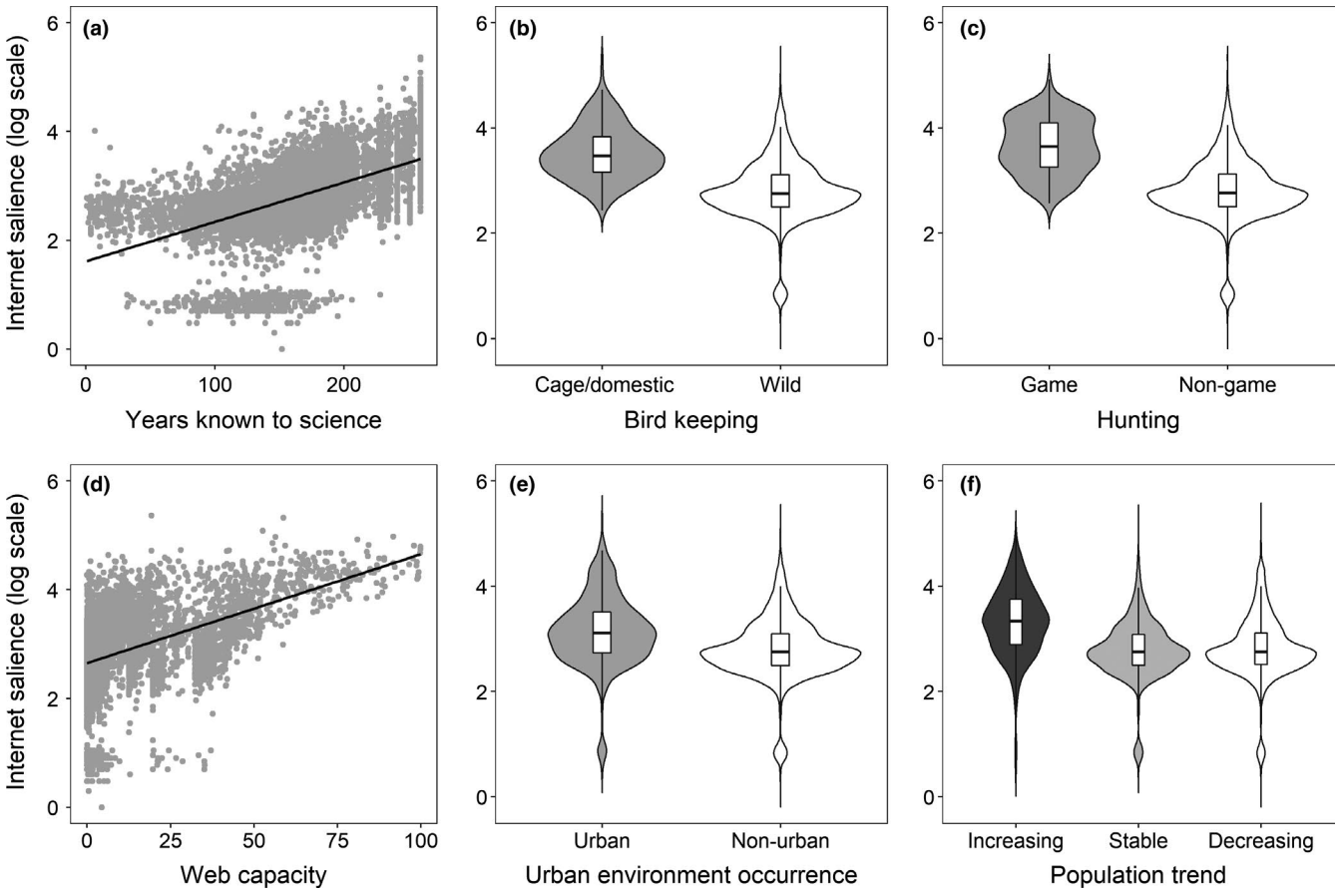

FIGURE 1 Effect of selected biocultural variables on internet salience of extant bird species: (a) years since scientifically described; (b) bird keeping; (c) hunting; (d) % global internet users in countries where species is present; (e) presence in urban environments; (f) population trend. Black lines in plots (a) and (d) illustrate the linear relationships suggested by our modelling results. In the remaining plots, boxes indicate median, first and third quartiles and the width of violin plots illustrates the density distribution of the data. Full description of variables and model results is given in Tables S3–S7

that are involved in non-scientific cultural practices. Finally, there are also biases associated with the geographies of the internet (Zook, 2006) – where web content is generated and by whom.

To understand some of the technical limitations of our approach, it is necessary to outline how internet search algorithms estimate the number of webpages (known as 'hit counts' or 'hits') that contain a search term. Contrary to popular belief, they do not compare the search term to every webpage on the global internet (Uyar, 2009). Rather, they only check webpages within their own index, an enormous database of information (rather like the index of a book) about webpages generated by specialized software known as web crawlers. Google's index is over 100,000,000 gigabytes and contains hundreds of billions of webpages (Google, 2019). Since each search engine algorithm compares the search query to its own index, no two search engines will produce the same estimate, since they index different proportions of the internet and use different algorithms to estimate matching webpages. Moreover, since the index is continually updated hit counts are likely to increase over time independent of cultural trends.

Another potential limitation of our metric relates to the scope of webpages retrieved by the search terms used. Because species scientific names were used as search terms, our metric is likely to be biased towards institutional and technical webpages (that are more likely to use scientific names) and, conversely, biased against more popular or traditional discourse about birds. Nevertheless, the strong observed correlation between vernacular names and common names on the internet and other digital corpora supports its use as a general measure of cultural representation on the internet (Correia et al., 2017; Jarić et al., 2016). Moreover, in order to control for the influence of taxonomic uncertainty (Correia et al., 2018), we included all known taxonomic synonyms when calculating our metric. It follows that, all else being equal, bird species with little cultural salience or only salient for people with low levels of engagement with the global internet will have correspondingly low representation on the global internet. This characteristic of our metric can be seen in its strong correlation with web capacity (the percentage of global internet users in the countries where the species is present – Figure 1d).

Furthermore, logic suggests that the relative ranking of search terms, in our case scientific names of bird species, is robust when compared to each other. In other words, species with a relatively high number of hits will rank highly across search engines and over time using the same search engine. This is because the algorithms work in a similar way and all the major search engines have all indexed a high proportion of the global internet (Uyar, 2009). Such relative comparability was clearly demonstrated by Thelwell (2008) who reported strong correlations between the hit count estimates of the Google, Yahoo and Live Search search engines. He recommended Google for obtaining the most accurate estimates – a recommendation we have followed. It is important to note that hit counts are sensitive to the number of words used in a query, with single word or single word units (e.g. multiple words in quotation marks) providing nearly double the hit count accuracy as compared to multi-word queries (without quotation marks; Uyar, 2009) – this is controlled in our study by the exclusive use of binomial names in quotes.

It is also important to note that the large size of datasets generated by culturomics approaches can pose a challenge for traditional statistical analyses based around statistical 'significance'. This is because when working with large datasets, such as those generated by culturomics methods, even small numerical differences are likely to be identified as statistically significant due to large sample sizes. For this reason, we follow recent recommendations (Amrhein, Greenland, & McShane, 2019) and structure the discussion of our results in terms of standardized effect sizes rather than around statistical significance.

In addition to uncertainties about the extent, geography and identity of the digital corpora being searched, researchers also do not know exactly how search data such as hit counts are generated. This is because the exact technical specifications of internet search algorithms are commercial secrets. Moreover, part of the success of search algorithms such as Google Search is related to its capacity to deliver highly personalized search results and, although this can be 'deactivated' prior to data capture, it is not known at what level of the internet architecture (e.g. Global, national) it is deactivated. To account for this uncertainty, we routed searches via a distributed network of proxy servers, but the effectiveness of this 'de-personalization' strategy is unknown. Furthermore, along with its index of webpages, Google's search algorithms are under continuous development. This further constrains replicability and creates an imperative to archive harvested data used in studies such as this on open access data repositories (Ladle et al., 2016).

Culturomics research is also subject to the epistemological challenges associated with the enormous variation in use of language (Ladle et al., 2016), some of which we address through the use of scientific names and taxonomic synonyms. However, while these strategies allow us to confidently identify which species are more represented on the global internet, our metric does not distinguish between positive and negative sentiment. Thus, a species could be highly represented due to frequent negative interactions with people (e.g. crop raiding, spreading disease, etc.) or a mixture of positive and negative interactions. Such information is clearly of great interest to conservationists but is particularly difficult to automatically assess due to the semantic complexity/diversity of languages and the frequent use of idiomatic language to describe feelings (Aggarwal, 2018). Even if automated sentiment analysis was effective, it would almost certainly be better applied to other digital corpora such as those derived from social media platforms (e.g. Twitter).

Our analysis was also constrained by the availability of input data for our model. For example, we complied initial lists of game and cage and aviary birds from the rather limited set of readily available sources. Other potentially fascinating datasets were either non-existent, unavailable or incomplete, including data on the aesthetic appeal of species, their popularity as zoo exhibits, emblems and mascots, and their presence in religious texts and children's books – all of which are likely to contribute to the cultural salience of a species. Interactions of people with representations of a species may also

positively influence webpage mentions of a species, creating a positive feedback and potentially increasing inequality in representation over time. This would be fascinating to investigate but requires good quality historical data on past representation.

Finally, a limitation that belongs to all forms of culturomic analysis is related to who is responsible for generating the data and where they live. Many rural and indigenous populations in developing countries are of great interest to conservation, but have limited internet access and lack visibility on digital corpora thereby placing them beyond the reach of culturomics research. In a similar vein, digital corpora are more complete for the major language groups (Funk & Rusowsky, 2014). These characteristics limit the potential of culturomics to the study of mass internet culture and culturomic analysis may not be applicable to all issues of conservation interest (Ladle et al., 2016). In the current study, we attempted to account for and explore the effects of differential internet access in our model by including variables representing: (a) web capacity of countries where the species is present (Figure 1d), and; (b) urban bird species (Figure 1e).It would be fascinating to assess the relative contribution to global internet salience from webpages hosted by domains outside of a species' country-level distribution. This would potentially allow conservationists to identify universally or regionally iconic species and to assess public knowledge and interest in countries where a species has become locally extirpated.

5.2 | Associations with explanatory variables

As outlined above, we hypothesize that the internet salience of a species should be strongly influenced by its cultural prominence, the latter delimited by a combination of its intrinsic traits and its interactions with human culture and psychology. Our supposition is supported by the strong correlations between our internet salience metric and many of the explanatory variables in our model. The observed association between global internet salience and the number of years since a species was first scientifically named is to be expected for two main factors. First, there were strong historical biases in the geographic distribution and traits of the bird species known to science at the birth of modern taxonomy (e.g. 1750–1850). Early taxonomists would presumably have worked on species that were widespread and conspicuous (Gaston & Blackburn, 1994), sent back from European voyages of discovery and colonies, and/or that had curiosity, aesthetic, narrative or utility value in relation to science, literature and culture. Second, there is a time-lag between scientifically naming a species and the wider cultural recognition of that species (Hunn, 1999).

Aviculture and bird keeping also appear have a strong association with internet representation (Figure 1b). This is unsurprising given that these activities are ancient, widespread and several species (e.g. *Serinus canaria* Linnaeus, 1758, *Agaporis fischeri* Reichenow, 1887) have domestic varieties that are globally popular (Birkhead & van Balen, 2005). The high global salience of game species undoubtedly reflects the widespread cultural legacy of hunting (Kellert, 1985) including, for many species, a history of extra-limital introductions (Long, 1981) that greatly expands their geographical distribution. Hunting is frequently intertwined with self-esteem, history and cultural identity of both modern and ancient societies (Loveridge, Reynolds, & Milner-Gulland, 2006). It should be noted that these values, along with the religious and spiritual values derived from species, are typically strongly dependent on long ('sedimented') histories of cultural engagement that are impossible to simply capture within a model. Nevertheless, it could equally be argued that the chronological order in which species were described provides a very broad measure of their cultural embeddedness.

The global prominence of bird species found in urban environments (Figure 1e) and those with expanding populations (Figure 1f) is also unsurprising given that both these characteristics should be associated with high levels of public familiarity especially given the widely distributed nature of many urban-associated bird species (Møller, 2009). Moreover, urban populations typically have greater access to the internet and are likely to contribute a greater proportion of web content generators. Interestingly, higher endangerment categories were also generally associated with higher cultural representation – although the relationship is less pronounced than some of the other variables. This association is presumably partly driven by the increased publicity given to these species and conservation-motivated practices of engagement. It may also reflect the value these species provide for education and research. Conservation research, especially, is orientated towards rare, endemic and endangered species, or species that endanger other species such as those that have been intentionally or accidentally introduced out of their native range. Once again, education and research value is likely to strongly overlap with other cultural values, with cultural salience being both a motivation and justification for further study.

5.3 | Potential applications

Internet-based metrics of the cultural salience of species have a range of potential applications and may prompt policy innovation. For example, they offer a means to measure, map and report stocks of cultural natural capital in an ecosystem or region in near real time, and could therefore contribute to the broader assessment of nature's contribution to people (Díaz et al., 2018). All things being equal, species with high cultural significance are more likely to be assets that can generate value in the form of collective identities, brands, life-quality and nature-based economies (Jepson et al., 2017).

Internet-based metrics of cultural salience of species may also provide information on the vulnerability of species to certain anthropogenic threats (e.g. habitat loss due to infrastructure development) because publics are more likely to rally in support of species they know and value (Brambilla et al., 2013; Macdonald et al., 2015). Cultural salience metrics could also be used to critically appraise the democratic accountability and transparency of policy including, for example, the degree to which contemporary biodiversity policy and funding meets public expectations of which species are conserved and 'cared for' or how nature-related policy serves recreational practices. Finally, cultural salience metrics could be used to inform

and monitor the effectiveness of strategies aimed to support Aichi Biodiversity Target 1, increasing public awareness of the value of biodiversity and actions that help to conserve it (Cooper et al., 2019).

6 | CONCLUSIONS

Developments in biodiversity informatics and environmental modelling systems have massively improved the ability of scientists to inform policy makers and markets on issues of environmental management and sustainability (Recknagel & Michener, 2017). However, biodiversity datasets have typically been narrowly focused on taxonomic and ecological data, leading Hardisty and Roberts (2013) to urge conservationists to adopt a more holistic, whole system approach that takes 'biodiversity science far beyond a collection of taxon names'. We would argue that our webpage-based metric of cultural salience of species could contribute to such a 'whole system' approach, as could a diverse range of other culturomics-derived metrics (Correia et al., 2019). Indeed, recent studies have demonstrated the utility of culturomic approaches for quantifying and monitoring trade in endangered species via information on social media sites (Di Minin, Fink, Tenkanen, & Hiippala, 2018; Di Minin, Tenkanen, & Toivonen, 2015) and the enormous potential of user generated data (e.g. Wikipedia page views) for tracking of taxonomic and temporal trends in public interest in nature (Mittermeier, Roll, Matthews, & Grenyer, 2019; Roll et al., 2016). A genuine 'whole system' approach would also need to incorporate data on human sentiment towards species and places. Enormous technical advances have already been made in this direction (Liu, 2015), and the increasing sophistication of machine-learning approaches to sentiment analysis of digital texts (Zhang, Wang, & Liu, 2018) suggests that robust automated quantification and tracking of human sentiment towards species will be feasible in the near future.

Despite these exciting advances, there is clearly still a long way to go to fully understand what types of cultural engagements are represented in different digital corpora, how to work with these corpora, and the biases involved (Ladle, Jepson, Correia, & Malhado, 2017). Nevertheless, in contrast to the continuing decline in human contact with the natural world (Soga & Gaston, 2016), digital representations of biodiversity (and human interactions with it) are rapidly increasing. When combined with the trend of increasing cross platform inter-operability (Kecskemeti, 2016), this is likely to further increase the power and reach of culturomic research and the next few years are likely to bring exciting new developments in this emerging field of conservation research (Ladle et al., 2016).

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CONFLICT OF INTEREST

The authors do not have any conflict of interest.

AUTHORS' CONTRIBUTIONS

R.J.L., P.J., R.A.C. and A.C.M.M. designed and implemented the study. R.A.C. collated and analysed the data. R.J.L. and P.J. wrote the manuscript.

DATA AVAILABILITY STATEMENT

Raw data associated with statistical analysis and Figure 1 are fully available in the Supporting Information and are also archived on Figshare: https://doi.org/10.6084/m9.figshare.7952522.v1 (Ladle, Jepson, Correia, & Malhado, 2019).

CODE AVAILABILITY

Code is not required to reproduce results, but can be requested from the authors.

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